

SUMMATIVE ASSESSMENT: EVALUATING LEARNING OUTCOMES IN THE FORM OF TERMINAL/FINAL EXAMINATION

Summative assessment is a fundamental component of the educational process, designed to evaluate student learning at the conclusion of an instructional period. Unlike formative assessment, which provides ongoing feedback during the learning process, summative assessment serves as a final measure of what students have learned. Typically administered at the end of a unit, course, or academic year. Summative assessment aims to determine the extent to which educational objectives have been achieved.

The primary purpose of summative assessment is to assess the overall effectiveness of instruction and learning. It provides a conclusive evaluation of student performance, often in the form of tests, final projects, or standardized exams. These assessments generate grades or scores that reflect a student's achievement in a given subject area over a specific period or time duration.

Summative assessment is often used to make critical decisions regarding student progression, certification, or placement in subsequent educational levels. Additionally, summative assessments provide valuable data that inform curriculum development and instructional strategies. By analyzing summative assessment results, educators can identify trends, strengths, and weaknesses within their instructional approaches, allowing for improvements in future teaching.

In conclusion, summative assessment plays a critical role in the educational process by providing a final evaluation of student learning. While it differs from formative assessment in its focus and application, it is an essential tool for measuring academic achievement. When balanced with formative assessments, summative assessments contribute to a well-rounded and effective approach to evaluating and supporting student learning.

National Curriculum of Pakistan 2022-2023
Assessment Framework Chemistry
Details of Content Area/ SLOs----- Grade-IX (SSC-I)

Domains	Content Area	SLO No.	SLO Description	Cognitive Level	Type of Assessment	Remarks	Time allocation (Periods 1Period =40 minutes)
A	Nature of Science in Chemistry	SLO: C-09-A-01	Define chemistry as the study of matter, its properties, composition and interactions with other matter and energy or study of Earth (solids), Air (gases), Sea (liquids) and sky (plasma) and their interaction with each other.	Knowledge	Summative	Question(s) will be asked in annual examination.	04
		SLO: C-09-A-02	Explain with examples that chemistry has many sub-fields and example, and interdisciplinary fields. (Some examples includes). Biochemistry. Medicinal Chemistry. Polymer Chemistry. Geochemistry. Environmental Chemistry. Analytical Chemistry. Physical Chemistry. Organic Chemistry. Inorganic Chemistry. Nuclear Chemistry. Astrochemistry).	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-A-03	Formulate examples of essential question(s) that are important for the branches of Chemistry. e.g. for Analytical Chemistry a question would be 'how can we accurately determine the chemical composition of a sample?')	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-A-04	Differentiate between 'science', 'technology' and 'engineering' by making reference to examples from the physical sciences. (Science is a process of exploring new knowledge methodically through observation and experiments, technology refers to the process of applying scientific knowledge in practical applications, for various purposes. Engineering is the application of knowledge in order to design, build and maintain a product or a process. that solves a problem and fulfills a need. Science provides the fundamental knowledge and understanding whole engineering applies that knowledge to develop practical solutions.	Understanding	Formative	Question(s) will not be asked in annual examination, however it will be a part of classroom teaching.	
B	Matter	SLO: C-09-B-01	Define matter as a substance having mass and occupying space.	Knowledge	Summative	Question(s) will be asked in annual examination	07
		SLO: C-09-B-02	State the distinguishing macroscopic properties of commonly observed states of solids, liquids and gasses in particular density, compressibility and fluidity.	Understanding	Summative	Question(s) will be asked in annual examination.	

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		SLO: C-09-B-03	Identify that state is a distinct form of matter (examples could include familiarity with plasma, intermediate states and exotic states e.g. BEC or liquid crystals).	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-04	Explain the allotropic forms of solids(some examples may include diamond, graphite, and fullerenes)	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-05	Explain the differences between elements, compounds and mixtures.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-06	Identify solutions, colloids, and suspensions as mixtures and give an example of each.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-07	Explain the effect of temperature on solubility and formation of unsaturated and saturated solutions.	Understanding	Summative	Question(s) will be asked in annual examination.	
	Atomic structure	SLO: C-09-B-08	Explain the structure of the atom as a central nucleus containing neutrons and protons surrounded by electrons in shells.	Understanding	Summative	Question(s) will be asked in annual examination.	13
		SLO: C-09-B-09	State that, orbits (shells) are energy levels of electrons and a larger shell implies higher energy and greater average distance from nucleus.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-10	State that electrons are quantum particles with probabilistic paths whose exact paths and locations cannot be mapped (with reference to the uncertainty principle).	Knowledge	Formative	Question(s) will not be asked in annual examination, however it will be a part of classroom teaching.	
		SLO: C-09-B-11	Explain that a nucleus is made up of protons and neutrons held together by strong nuclear force.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-12	Explain that an atomic model is an aid to understand the structure of an atom.	Understanding	Formative	Question(s) will not be asked in annual examination, however it will be a part of classroom teaching	
		SLO: C-09-B-13	State the relative charge and relative masses of a subatomic particles (an electron, proton and neutron).	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-14	Interpret the relationship between a subatomic particle, their mass and charge	Understanding	Summative	Question(s) will be asked in annual examination.	

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		SLO: C-09-B-15	Illustrate the path that positively and negatively charged particles would take under the influence of a uniform electric field.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-16	Define proton number/atomic number as the number of protons in the nucleus of an atom.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-17	Explain that the proton number is unique to each element and used to arrange elements in periodic table.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-18	State that radioactivity can change the proton number and alter an atom's identity.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-19	Define nucleon number/atomic mass as sum of number of protons and neutrons in the nucleus of an atom.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-20	Define isotopes as different atoms of the same element that have same number of protons but different neutrons	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-21	State that isotopes can affect molecular mass but not chemical properties of an atom.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-22	Determine the number of protons and neutrons of different isotopes.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-23	Define relative atomic mass as the average mass of isotopes of an element compared to 1/12th of mass of an atom of Carbon-12.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-24	State that isotopes can exhibit radioactivity.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-25	Discuss the importance of isotopes using carbon dating and medical imaging as examples.	Application	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-26	Describe the formation of positive (cation) and negative (anion) ions from atoms.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-27	Interpret and use the symbols for atoms and ions.	Knowledge	Summative	Question(s) will be asked in annual examination.	

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		SLO: C-09-B-28	Calculate relative atomic mass of an element from relative masses and abundance of isotopes.	Application	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-29	Calculate relative atomic mass of an element from relative masses and abundance of all stable isotopes.	Application	Summative	Question(s) will be asked in annual examination.	
	Chemical Bonding	SLO: C-09-B-30	Describe that noble gas electronic configuration, octet and duplet rules help predict chemical properties of main group elements.	Understanding	Summative	Question(s) will be asked in annual examination.	09
		SLO: C-09-B-31	Compare between the formation of cations and anions.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-32	Account for the electropositive and electronegative nature of metals and non-metals.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-33	Define ionic, covalent, coordinate covalent and metallic bonds.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-34	Differentiate between ionic compounds and covalent compounds.(The following points need to be included in the respective, a. ionic bond as strong electrostatic attraction between oppositely charged ions b. Covalent bond as strong electrostatic attraction between shared electrons and two nuclei. C. Metallic bond as strong electrostatic attraction between cloud/sea of delocalized electrons and positively charged cations).	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-35	Explain the properties of compounds in terms of bonding and structure.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-36	Compare uses and properties of materials such as strength and conductivity as determined by the type of chemical bond present between their atoms.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-37	Interpret the strength of forces of attraction and their impact on melting and boiling points of ionic and covalent compounds.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-38	Justify the availability of free charged particles (electrons or ions) for conduction of electricity in ionic compounds (solid in molten) covalent compounds and metallic bonds.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-39	Recognize that some substances can ionize when dissolved in water.(e.g. acids dissolves in water and conduct electricity).	Knowledge	Summative	Question(s) will be asked in annual examination.	

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		SLO: C-09-B-40	Justify the suitability of usage of graphite, diamond and metals for industrial purposes. Some examples may include : a. graphite as lubricant or an electrode. b. Diamond in cutting tools. c. Metals for wires, and sheets).	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-41	Draw the structure of ionic and covalent compounds along with their formation. (some examples can include: a. ionic bonds in binary compounds such as NaBr, NaF, CaCl ₂ using dot-and cross diagrams and Lewis dot structures b. simple molecules including H ₂ , Cl ₂ , O ₂ , N ₂ , H ₂ O, CH ₄ , NH ₃ , HCl, CH ₃ OH, C ₂ H ₄ , CO ₂ , HCN, and similar molecules using dot-and- cross diagrams and Lewis dot structures).	Application	Summative	Question(s) will be asked in annual examination.	
	Stoichiometry	SLO: C-09-B-42	State the formulae of common elements and compounds.	Knowledge	Summative	Question(s) will be asked in annual examination.	12
		SLO: C-09-B-43	Define molecular formula of a compound as the number and type of different atoms in one molecule.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-44	Define empirical formula of a compound as the simplest whole number ratio of different atoms in a molecule.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-45	Deduce the formula and name of a binary ionic compounds from ions given relevant information.	Application	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-46	Deduce the formula of a molecular substance from the given structure of molecules.	Application	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-47	Use the relationship. Amount of substance = mass / molar mass to calculate number of moles, mass, molar mass, relative mass (atomic/molecular/formula) and number of particles.	Application	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-48	Define mole as amount of substance containing Avogadro's number (6.02x10 ²³) of particles.	Knowledge	Summative	Question(s) will be asked in annual examination.	

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		SLO: C-09-B-49	Explain the relationship between a mole and Avogadro's constant.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-50	Construct chemical equations and ionic equations to show reactants forming products, including state symbols.	Application	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-51	Deduce the symbol equation with state symbols for a chemical reaction given relevant information.	Application	Summative	Question(s) will be asked in annual examination.	
	Electrochemistry	SLO: C-09-B-52	Define redox reactions as simultaneous oxidation and reduction in terms of oxygen, hydrogen, electrons and changes in oxidation state.	Knowledge	Summative	Question(s) will be asked in annual examination.	12
		SLO: C-09-B-53	Use roman numerals to indicate oxidation number of an element in a compound.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-54	Identify oxidizing and reducing agents in a redox reaction.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-55	Recognize that the oxidation number of elements in their free state is zero.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-56	Derive the formula of ionic compounds from ionic charges and oxidation numbers.	Application	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-57	Identify that the oxidation number of monatomic ion is the same as the charge on the ion.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-58	Explain that the sum of the oxidation numbers in a neutral compound is zero.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-59	Explain that the sum of the oxidation numbers in an ion is equal to the charge on the ion.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-60	Identify redox reactions by the colour changes involved when using acidified aqueous potassium manganate (VII) to (II) or aqueous potassium iodide.	Understanding	Summative	Question(s) will be asked in annual examination.	

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		SLO: C-09-B-61	Define corrosion and discuss methods to prevent it. (Some examples may include barrier method such as using paint galvanizing, electroplating; sacrificial protection such as using magnesium blocks in ships).	Understanding	Summative	Question(s) will be asked in annual examination.	
	Energetics	SLO: C-09-B-62	Explain the idea of a chemical system and its connection with its surroundings influences energy transfer during a chemical reaction.	Knowledge	Summative	Question(s) will be asked in annual examination.	07
		SLO: C-09-B-63	Differentiate between exothermic and endothermic reactions by giving examples.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-64	State that thermal energy is called enthalpy change and recognize its sign as negative for exothermic and positive for endothermic reactions.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-65	Define activation energy as the minimum energy that colliding particles must have for a successful collision.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-66	Explain that activation energy depends on reaction pathway which can be changed using catalysts or enzyme (detailed pathways not required).	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-67	Draw, label and interpret reaction pathway diagram for exothermic and endothermic reaction which includes enthalpy change, activation energy (uncatalyzed and catalyzed), reactants and products.	Application	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-68	Recognize that bond breaking is endothermic and bond making is exothermic processes.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-69	Explain that enthalpy change is sum of energies absorbed and released in bond breaking and bond forming.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-70	Calculate enthalpy change of a reaction given bond energy values.	Application	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-71	Explain how respiration (aerobic and anaerobic), an exothermic process, provides energy for biological systems and lipids as reserve stores of energy.	Understanding	Summative	Question(s) will be asked in annual examination.	
	Chemical Equilibrium	SLO: C-09-B-72	Recognize that reversible reaction are shown by symbol ($\rightleftharpoons$) and may not go to completion.	Understanding	Summative	Question(s) will be asked in annual examination.	05

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		SLO: C-09-B-73	Describe how changing the physical conditions of a chemical equilibrium system can redirect reversible reactions. a. effect of heat on hydrated compounds b. addition of water to anhydrous substances in particular copper(II) sulfate and cobalt (II) chloride	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-74	State that reversible reaction can achieve equilibrium in a closed system when rate of forward and backward reactions are equal.	Knowledge	Summative	Question(s) will be asked in annual examination.	
	Acids, Bases Chemistry and pH	SLO: C-09-B-75	Define Bronsted-Lowry acids as proton donors and Bronsted-Lowry bases as proton acceptors.	Knowledge	Summative	Question(s) will be asked in annual examination.	09
		SLO: C-09-B-76	Recognize that aqueous solutions of acids contain H ⁺ ions and aqueous solutions of alkalis contain OH ⁻ ions.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-77	Define a strong acid and bases as an acid or base that completely dissociates in aqueous solution and weak acid and base that partially dissociates in aqueous solution. (Some examples include: Student writing symbol equations to show these for hydrochloric acid, sulphuric acid, nitric acid, and ethanoic acid.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-78	Formulate dissociation equations for an acid or base in aqueous solution.	Application	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-79	Recognize that bases are oxides or hydroxides of metals and that alkalis are water-soluble bases.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-80	Describe the characteristic properties of acids in terms of their reactions with metals, bases and carbonates.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-81	Identify the characteristic properties of bases in terms of their reactions with acids and ammonium salts.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-82	Define acid rain.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-B-83	Discuss effects of acid rain and relate them with properties of acids.	Application	Summative	Question(s) will be asked in annual examination.	

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C	Periodic Table & Periodicity	SLO: C-09-C-01	Define the periodic table as an arrangement of elements in periods and groups in order of increasing proton number / atomic number.	Knowledge	Summative	Question(s) will be asked in annual examination.	20
		SLO: C-09-C-02	Identify the group or period or block of an element using its electronic configuration (only the idea of subshells related to the blocks can be introduced).	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-C-03	Explain the relationship between group number and the charge of ions formed from elements in the group in terms of their outermost shells.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-C-04	Explain similarities in the chemical properties of elements in the same group in terms of their electronic configuration.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-C-05	Identify trends in group and periods, given information about the elements, including trends for atomic radius, electron affinity, electronegativity, ionization energy, metallic character, reactivity and density.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-C-06	Use terms alkali metals, alkaline earth metals, halogens, noble gases, transition metals, lanthanides and actinides in reference to the periodic table.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-C-07	Predict the characteristic properties of an element in a given group by using knowledge of chemical periodicity.	Application	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-C-08	Deduce the nature, possible position in the Periodic Table and the identity of unknown elements from given information about their physical and chemical properties.	Application	Summative	Question(s) will be asked in annual examination.	
	Group Properties and Elements	SLO: C-09-C-09	Group I Properties Define Group I, Alkali metals as relatively soft metals with general trends down the group limited to decreasing melting point, increasing density and increasing reactivity.	Knowledge	Summative	Question(s) will be asked in annual examination.	10
		SLO: C-09-C-10	Predict properties of other elements in group I, given information about the elements.	Application	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-C-11	Predict properties of elements in group I in order of reactivity given relevant information.	Application	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-C-12	Group VII Properties Define Group-VII, halogens as diatomic non-metals with general trends limited to increasing density, and decreasing reactivity.	Knowledge	Summative	Question(s) will be asked in annual examination.	

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		SLO: C-09-C-13	Identify the appearance of halogens as fluorine as pale yellow gas, chlorine as yellow-green gas, bromine as red-brown liquid, iodine as grey black solid.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-C-14	Explain the displacement reactions of halogens with other halide ions and also as reducing agents.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-C-15	Predict the properties of elements in group VII, given information about the elements.	Application	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-C-16	Analyze the relative thermal stabilities of the hydrogen halides and explain these in terms of bond strengths.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-C-17	Transition elements Describe Transition elements as metals that: have high densities, high melting points, variable oxidation numbers, form colored compounds and act as catalysts for industrial purposes. (some examples include catalysts being used are the Haber process, catalytic converters, Contact process and manufacturing of margarine).	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-C-18	Noble gases Define the Group 18 noble gases as unreactive, monatomic gases.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-C-19	Explain this (Nobel gases) in terms of electronic configuration.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-C-20	Properties of metals Compare the general physical properties of metals and non-metals. (Specifically in terms of, a. thermal conductivity, b. electrical conductivity, c. malleability and ductility. d. melting points and boiling points)	Understanding	Summative	Question(s) will be asked in annual examination.	
D	Atmosphere	SLO:C-09-D-01	State that composition of clean, dry air is approximately 78% Nitrogen N ₂ , 21% Oxygen O ₂ , and the remainder as a mixture of noble gases and carbon dioxide CO ₂ .	Knowledge	Summative	Question(s) will be asked in annual examination.	07

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		SLO:C-09-D-02	State the major sources of air pollutants. (Some examples include: a. Carbon dioxide from the complete combustion of carbon-containing fuels, b. Carbon monoxide and particulates from the incomplete combustion of carbon-containing fuels, c. methane from the decomposition of vegetation and waste gasses from digestion in animals. d. Oxides of nitrogen from car engines, e. Sulfur dioxide from the combustion of fossil fuels which contain sulfur compounds f. Ground level ozone from reactions of oxides of nitrogen, from car engines, and volatile organic compounds, in presence of light).	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO:C-09-D-03	State the adverse effects of air pollutants (Some examples include: a. carbon dioxide: higher levels of carbon dioxide leading to increased global warming, which leads to climate change b. carbon monoxide: toxic gas c. particulates: increased risk of respiratory problems and cancer d. methane: higher levels of methane leading to increased global warming, which leads to climate change e. oxides of nitrogen: acid rain, photochemical smog and respiratory problems, f. sulfur dioxide: acid rain and haze).	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO:C-09-D-04	Explain how the greenhouse gasses carbon dioxide and methane cause global warming? (Some examples include : a. The absorption, reflection and emission of thermal energy, b. Reducing thermal energy loss to space).	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO:C-09-D-05	Describe the role of sulfur in the formation of acid rain and its impact on the environment.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO:C-09-D-06	Describe the strategies to reduce the effects of major environmental issues. (Some examples include: a. climate change: planting trees, reduction in livestock farming, decreasing use of fossil fuels, increasing use of hydrogen and renewable energy, e.g. wind, solar. b. acid rain: use of catalytic converters in vehicles, reducing emissions of sulfur dioxide by using low sulfur fuels and flue gas desulfurization with calcium oxide).	Application	Summative	Question(s) will be asked in annual examination.	
		SLO:C-09-D-07	Describe the role of NO and NO ₂ in the formation of acid rain, both directly and through their catalytic role in the oxidation of atmospheric sulfur dioxide.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO:C-09-D-08	Explain how oxides of nitrogen form in car engines and describe their removal by catalytic converters, e.g. $\text{CO} + 2\text{NO} \longrightarrow 2\text{CO} + \text{N}_2$	Understanding	Summative	Question(s) will be asked in annual examination.	

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		SLO:C-09-D-09	Define photosynthesis as the reaction between carbon dioxide and water to produce glucose and oxygen in the presence of chlorophyll and using, energy from light.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO:C-09-D-10	Analyze how to use tools to reduce personal exposure to harmful pollutants. (Some examples include the usage of masks, air quality indices and CO detectors).	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO:C-09-D-11	Identify high risk situations in life including those where long-term exposure to these pollutants can lead to respiratory issues and reduction in quality and longevity of life.	Understanding	Summative	Question(s) will be asked in annual examination.	
	Environmental Chemistry-Water	SLO: C-09-D-12	Investigate chemical tests for the presence of water using anhydrous copper (II) sulfate.	Understanding	Summative for PBA (Practical Based Assessment)	Lab work-Question(s) will be asked in PBA	06
		SLO: C-09-D-13	Explain how to test the purity of water using melting point and boiling point.	Understanding	Summative for PBA	Lab work-Question(s) will be asked in PBA	
		SLO: C-09-D-14	Distinguish between Distilled water and tap water with their applications in practical chemistry.	Understanding	Summative for PBA	Lab work-Question(s) will be asked in PBA	
		SLO: C-09-D-15	State that water from natural sources may contain useful and harmful substances. (Some examples include: a. dissolved oxygen b. metal compounds. c. plastics d. sewage. e. harmful microbes. f. nitrates from fertilizers. g. phosphates from fertilizers and detergents)	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-D-16	Recognize that some naturally occurring substances in water are beneficial. (Some examples include, : a. Dissolved oxygen for aquatic life. b. Some metal compounds provide essential minerals for life.)	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-D-17	Recognize that some naturally occurring substances in water are potentially harmful. (Some examples include: a. some metal compounds that are toxic, b. some plastics that harm aquatic life. c. sewage that contains harmful microbes	Understanding	Summative	Question(s) will be asked in annual examination.	

Domains	Content Area	SLO No.	SLO Description	Cognitive Level	Type of Assessment	Remarks	Time allocation (Periods 1Period =40 minutes)
			which cause diseases. d. nitrates and phosphates that lead to deoxygenation of water and damage to aquatic life. Details of the eutrophication process are not required).				
		SLO: C-09-D-18	Explain the treatment of the domestic water supply. (Some examples of this includes. a. Sedimentation and filtration to remove solids. b. Use of carbon to remove tastes and odors. c. chlorination to kill microbes).	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-D-19	Describe various water-borne diseases and the steps that can be taken to avoid them.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-D-20	Identify the negative effects of water pollutants on life and the ways to avoid them.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-D-21	Explain water scarcity as an important issue faced by Pakistan and the ways in which it can be resolved.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-D-22	Fertilizers State that urea, ammonium salts and nitrates are used as fertilizers.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-D-23	Explain the use of NPK fertilizers to provide the elements nitrogen, phosphorus and potassium for improved plant growth.	Understanding	Summative	Question(s) will be asked in annual examination.	
E	Organic Chemistry	SLO: C-09-E-01	Describe organic molecules as either straight-chained, branched or cyclic.	Understanding	Summative	Question(s) will be asked in annual examination.	12
		SLO: C-09-E-02	State that a structural formula is an unambiguous description of the way the atoms in a molecule are arranged, including $\text{CH}_2=\text{CH}_2$, $\text{CH}_3\text{CH}_2\text{OH}$, $\text{CH}_3\text{COOCH}_3$.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-E-03	Identify and draw structural formulae for molecules.	Application	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-E-04	Interpret general formulae of compounds in the same homologous series including alkanes, alkenes, alkynes, alcohols and carboxylic acids.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-E-05	Define structural isomers as compounds with the same molecular formula, but different structural formulae, including C_4H_{10} as $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$; and C_4H_8 as $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$ and $\text{CH}_3\text{CH}=\text{CHCH}_3$	Understanding	Summative	Question(s) will be asked in annual examination.	

Domains	Content Area	SLO No.	SLO Description	Cognitive Level	Type of Assessment	Remarks	Time allocation (Periods 1Period =40 minutes)
		SLO: C-09-E-06	Identify a functional group as an atom or group of atoms that determine the chemical properties of a homologous series, including alcohols, aldehydes, ketones, phenols, carboxylic acids, amine, esters, and amide.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-E-07	Describe the general characteristics of a homologous series (These can include: a. Having the same functional group. b. Having the same general formula. c. Differing from one member to the next by a $\text{—CH}_2\text{—}$ unit. d. displaying a trend in physical properties. e. Sharing similar chemical properties).	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-E-08	State that a saturated compound has molecules in which all carbon-carbon bonds are single bonds.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-E-09	State that an unsaturated compound has molecules in which one or more carbon—carbon bonds are not single bonds.	Knowledge	Summative	Question(s) will be asked in annual examination.	
	Hydrocarbons	SLO: C-09-E-10	State that the bonding in alkanes is single covalent and that alkanes are saturated hydrocarbons.	Knowledge	Summative	Question(s) will be asked in annual examination.	07
		SLO: C-09-E-11	Describe the properties of alkanes as being generally unreactive, except in terms of combustion and substitution by chlorine.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-E-12	State that in a substitution reaction one atom or group of atoms is replaced by another atom or group of atoms.	Knowledge	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-E-13	Describe the substitution reaction of alkanes with chlorine as a photochemical reaction, and draw the structural or displayed formulae of the products, limited to monosubstitution.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-E-14	Describe, using symbol equations, preparation of alkanes from cracking of larger hydrocarbons, hydrogenation of alkenes and alkynes, and reduction of alkyl halides.	Understanding	Summative	Question(s) will be asked in annual examination.	
	Biochemistry	SLO: C-09-E-15	Explain the importance and basics of nutrition and healthy eating.	Understanding	Summative	Question(s) will be asked in annual examination.	05
		SLO: C-09-E-16	Recognize the main biomolecules; carbohydrates, proteins, lipids and nucleic acids, their sources, along with the required daily intake for young adults.	Understanding	Summative	Question(s) will be asked in annual examination.	
		SLO: C-09-E-17	Identify carbohydrates as a source of energy.	Understanding	Summative	Question(s) will be asked in annual examination.	